Worksheet 4: Equations, Functions and Systems

Write each answer in the space provided and show your working.

Part A: Equations and Inequalities

The common denominator of $\frac{1}{5}$ and $\frac{1}{2}$ is their product when the numbers share no factor. If a pump fills $\frac{1}{11}$ of a pool each hour, it needs 11 hours to fill the whole pool. Because $\frac{9x}{10}$ equals $\frac{9}{10}x$, the factor can be moved in front of the letter. The probability of drawing a red card is $\frac{3}{5}$ when 3 of the 5 cards are red.

A tank that is $\frac{5}{7}$ full holds 10 litres, so the whole tank holds $\frac{7}{5} \cdot 10$ litres. The reciprocal of $\frac{5}{7}$ is obtained by swapping the numerator and the denominator. The mean of 18 equal parts of a total T is $\frac{T}{18}$. The expression $\frac{2}{5}x + 5$ is linear, and its slope is $\frac{2}{5}$.

1. $f(x) = 2x^3 - 1x$. Find $f'(x)$ and the points where the tangent line is horizontal.

2. $\frac{5}{x} = \frac{7}{6}$. Solve the proportion for x.

3. $y = \frac{5x+9}{x-15}$. State the domain and the vertical asymptote of this function.

4. $\int_0^2 (5x + 9)\,dx$. Evaluate the definite integral.

5. $\sum_{k=1}^{8} (2k + 5)$. Evaluate the sum using the formulas for $\sum k$.

6. Simplify $\frac{x^2-25}{x+5}$ for $x \neq -5$.

7. $\log_3(x + 9) = 3$. Find x and state the domain of the logarithm.

8. $\frac{9}{x-1} + \frac{5}{x+1} = 5$. Solve for x, giving exact values.

9. $|x - 3| < 2$. Write the solution set in interval notation.

10. $0 = 18x^2 - 19x + 5$. Solve the equation for x by factoring.

11. $\frac{x}{9} + \frac{y}{9} = 1$. Find the points where the line crosses each axis.

12. Solve $\frac{x}{11} - \frac{x}{12} = 9$ for x and check your answer by substitution.

13. $3x + 15 \leq 5x - 1$. Solve the inequality and show the solution on a number line.

14. $y = 2x - 9$. Find the x-intercept of the line and state its slope.

15. $\sqrt{x + 13} = 7$. Solve the equation and check the answer for extraneous solutions.

16. Add $\frac{11}{2} + \frac{5}{4}$ and write the result as a single fraction in lowest terms.

Part B: Systems of Equations

A speed of $\frac{7}{9}$ metres per second is the same as 7 metres every 9 seconds. Writing the ratio of the sides as $\frac{3}{8}$ shows that the triangles are similar. Dividing both sides by 3 gives $x = \frac{5}{3}$. Multiplying $\frac{x}{2}$ by 2 removes the denominator and leaves the single letter x.

To compare $\frac{7}{9}$ with $\frac{11}{3}$, cross-multiply and compare the two products. A speed of $\frac{9}{3}$ metres per second is the same as 9 metres every 3 seconds. The probability of drawing a red card is $\frac{9}{11}$ when 9 of the 11 cards are red. A tank that is $\frac{11}{14}$ full holds 16 litres, so the whole tank holds $\frac{14}{11} \cdot 16$ litres.

17. Solve the system of equations by elimination:

$$\begin{cases} -9x + 3y = 33 \\ 2x + 3y = 0 \end{cases}$$

18. Find every pair of numbers that satisfies the system:

$$\begin{cases} 3x + 7y = 3 \\ 9x + 3y = 9 \end{cases}$$

19. Use substitution to solve the following system:

$$\begin{cases} -9x + 11y = 42 \\ -3x + 2y = 9 \end{cases}$$

20. Determine the values of the unknowns in the system:

$$\begin{cases} 11y - 3z = 22 \\ -7x + 7y = -14 \\ 11x + 11y + 3z = 66 \end{cases}$$

21. Solve the system of equations by elimination:

$$\begin{cases} 5x + 3y = -24 \\ -5x + 5y = 0 \end{cases}$$

22. Find every pair of numbers that satisfies the system:

$$\begin{cases} 9x + 7y = 57 \\ -5x + 11y = 13 \end{cases}$$

23. Use substitution to solve the following system:

$$\begin{cases} -3x + 9y = -12 \\ -7x + 2y = -28 \end{cases}$$

24. Determine the values of the unknowns in the system:

$$\begin{cases} 9x + 2y = -7 \\ -11x - 5y + 2z = 8 \\ -2y + 3z = 1 \end{cases}$$

25. Solve the system of equations by elimination:

$$\begin{cases} 3x - 3y = -3 \\ 7x + 11y = 29 \end{cases}$$

26. Find every pair of numbers that satisfies the system:

$$\begin{cases} 7x - 3y = -22 \\ -11x + 9y = 56 \end{cases}$$

27. Use substitution to solve the following system:

$$\begin{cases} -5x + 5y = -25 \\ 9x + 2y = 23 \end{cases}$$

28. Determine the values of the unknowns in the system:

$$\begin{cases} 11x + 11y + 3z = 9 \\ 9x + 9y + 3z = 9 \\ 3x - 5z = -12 \end{cases}$$

Part C: Functions Defined by Cases

The mean of 4 equal parts of a total T is $\frac{T}{4}$. The average rate of change over the interval is $\frac{f(7)-f(3)}{7-3}$. To compare $\frac{5}{8}$ with $\frac{9}{5}$, cross-multiply and compare the two products. Because $\frac{7x}{9}$ equals $\frac{7}{9}x$, the factor can be moved in front of the letter.

29. A function is defined in pieces:

$$g(t) = \begin{cases} 9t & \text{for } 0 \le t \le 5 \\ 45 & \text{for } 5 < t \le 10 \\ 45 - 5(t - 10) & \text{for } t > 10 \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

30. A function is given by cases:

$$u(x) = \begin{cases} 2x - 13 & \text{if } x > 13 \text{ and } x \neq 20 \\ 2 & \text{if } x = 20 \\ 0 & \text{elsewhere} \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

31. A function is defined in pieces:

$$s(n) = \begin{cases} 11 & \text{if } n \text{ is even} \\ -11 & \text{if } n \text{ is odd} \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

32. A function is given by cases:

$$P(w) = \begin{cases} 9 & \text{if } 0 < w \leq 3 \\ 9 + 3(w - 3) & \text{if } 3 < w \leq 8 \\ \text{no service} & \text{otherwise} \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

33. A function is defined in pieces:

$$s(n) = \begin{cases} 5 & \text{if } n \text{ is even} \\ -11 & \text{if } n \text{ is odd} \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

34. A function is given by cases:

$$s(n) = \begin{cases} 7 & \text{if } n \text{ is even} \\ -3 & \text{if } n \text{ is odd} \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

35. A function is defined in pieces:

$$u(x) = \begin{cases} 5x - 7 & \text{if } x > 13 \text{ and } x \neq 20 \\ 5 & \text{if } x = 20 \\ 0 & \text{elsewhere} \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

36. A function is given by cases:

$$P(w) = \begin{cases} 5 & \text{if } 0 < w \leq 13 \\ 5 + 2(w - 13) & \text{if } 13 < w \leq 22 \\ \text{no service} & \text{otherwise} \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

37. A function is defined in pieces:

$$g(t) = \begin{cases} 11t & \text{for } 0 \leq t \leq 1 \\ 11 & \text{for } 1 < t \leq 12 \\ 11 - 3(t - 12) & \text{for } t > 12 \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

38. A function is given by cases:

$$g(t) = \begin{cases} 9t & \text{for } 0 \leq t \leq 15 \\ 135 & \text{for } 15 < t \leq 20 \\ 135 - 11(t - 20) & \text{for } t > 20 \end{cases}$$

Evaluate the function at three inputs of your choice and describe its graph.

Part D: Mixed Review

The common denominator of $\frac{1}{9}$ and $\frac{1}{3}$ is their product when the numbers share no factor. The reciprocal of $\frac{5}{6}$ is obtained by swapping the numerator and the denominator. Dividing both sides by 5 gives $x = \frac{7}{5}$. On a map drawn to scale $\frac{1}{1100}$, one centimetre stands for 1100 centimetres on the ground.

The expression $\frac{11}{13}x + 3$ is linear, and its slope is $\frac{11}{13}$. Multiplying $\frac{x}{7}$ by 7 removes the denominator and leaves the single letter x. The average rate of change over the interval is $\frac{f(5)-f(3)}{5-3}$. A tank that is $\frac{3}{8}$ full holds 12 litres, so the whole tank holds $\frac{8}{3} \cdot 12$ litres.

On a map drawn to scale $\frac{1}{700}$, one centimetre stands for 700 centimetres on the ground. Two pipes together fill $\frac{1}{9} + \frac{1}{5}$ of the pool in one hour. Because $\frac{7x}{10}$ equals $\frac{7}{10}x$, the factor can be moved in front of the letter. The mean of 10 equal parts of a total T is $\frac{T}{10}$.

39. $a_n = 1 + (n-1) \cdot 5$. Find the 20th term of this arithmetic sequence.

40. $\sin\theta = \frac{7}{12}$. Find $\cos\theta$ for an acute angle θ.

41. $(x-5)^2 + (y+13)^2 = 36$. State the centre and the radius of this circle.

42. $\frac{d}{dx}\left(7x^3 - 15x^2\right)$. Evaluate this derivative at $x = 2$.

43. A line passes through $(15, 3)$ and $(25, 9)$. Compute the slope $\frac{9-3}{25-15}$ and write the line in point-slope form.

44. Rewrite $\frac{9x+9}{6}$ as the sum of two fractions and simplify each one.

45. $5^{x-1} = 25$. Solve for x by writing both sides with the same base.

46. $f(x) = 2x^3 - 1x$. Find $f'(x)$ and the points where the tangent line is horizontal.

47. $\frac{2}{x} = \frac{15}{5}$. Solve the proportion for x.

48. $y = \frac{2x+3}{x-3}$. State the domain and the vertical asymptote of this function.

49. Expand $(x+2)^2 - (x-15)^2$ and collect like terms.

50. Let $f(x) = 11x^2 - 5$. Compute $f(3) - f(-3)$ and explain why the answer is what you would expect.

51. A recipe uses $\frac{3}{8}$ cup of sugar for each batch. How much sugar is needed for 9 batches?

52. $\int_0^4 (3x + 3)\,dx$. Evaluate the definite integral.

53. $\sum_{k=1}^{12} (11k + 13)$. Evaluate the sum using the formulas for $\sum k$.

54. $\log_3(x+2) = 2$. Find x and state the domain of the logarithm.

55. $\frac{2}{x-1} + \frac{2}{x+1} = 1$. Solve for x, giving exact values.

56. $|x - 5| < 9$. Write the solution set in interval notation.